FASTQ Quality-Control Triage — Summary Report

Sample deliverable · 4-sample run

Peter Ristow, PhD — NGS Bioinformatics

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Executive summary. Four samples were triaged for data-quality problems before analysis. **3 of 4** had issues — adapter read-through, low-quality tails, and an overrepresented/contaminant sequence — all **diagnosed and fixed**, with the remaining sample passing clean. Every read removed has a documented reason; cleaning was deliberately conservative to avoid introducing bias.

Diagnosis & fix (before → after)

sample	problem	diagnosed	Q30 before→after	reads removed
S1_clean	none (baseline)	clean	100% → 100%	0.0%
S2_adapter	adapter read-through	adapters	100% → 100%	0.0%
S3_lowqual	low-quality 3′ tails	low quality, duplication/contaminant	27% → 27%	60.5%
S4_overrep	overrepresented sequence	adapters, duplication/contaminant	100% → 100%	44.9%

What this means. All four samples are now usable. The value is the *judgment* — reading the QC correctly, fixing what's fixable, and not over-trimming — not running the tools. Because the defects were known, the diagnosis is provably correct.

Caveats. Cross-species contamination (vs simple duplication) needs a screen such as Kraken2; a suspected sample swap is diagnosed via genotype/sex concordance, not read QC.

Reproducibility. The full triage re-runs from a clean clone in minutes and is verified by continuous integration. Code and live report: `github.com/pgristow/qc-rescue-fastq-triage`.

Prepared by Peter Ristow, PhD — NGS bioinformatics specialist. Production engagements run an automated triage across your whole run under NDA.